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ALTEN
Software Engineer – Fareham

Dear Hiring Manager,

Your Software Engineer role in Fareham needs someone who can build real-time C++ systems close to hardware and sensors, and keep them working once they're actually out in the field. That's what I've spent the last several years doing, most recently on industrial perception systems running 24/7, where a missed edge case meant someone driving out to the site.

At LivNSense I built asynchronous C++17 perception pipelines on Intel edge hardware, overlapping pre-processing, inference and post-processing to push throughput from 5 to 15 FPS while keeping real-time performance deterministic. I used Intel VTune to find the actual bottlenecks instead of guessing, and kept the refactor test-first with gtest so I'd know the moment something broke. It's the same kind of backend C++ work your listing is asking for, tuned and validated against real constraints.

My MSc dissertation built a real-time SLAM system in C++17 and ROS2, fusing LiDAR, depth-camera and radiological sensor data on a mobile robot, and at Esbaar I built an activity-classification workstream across 91 sensor channels alongside DeepStream pipelines processing multi-camera feeds, running continuously in a harsh desert environment. None of those sensors agree with each other cleanly, so a lot of the work is just getting the software to make a sensible call anyway.

At Esbaar I also hardened the deployed services themselves: Dockerised microservices, health checks, safe restarts, which cut live pipeline crashes by 67% and let the team recover without sending someone out. I led the field validation that tracked down failures under real lighting and motion conditions, and built the dashboard operators used to check live feeds without pulling in an engineer.

I'd like to talk through how this experience fits what your Fareham team is building. Thank you for considering my application, and I look forward to hearing from you.

Kind regards,
Dexter Fernandes